

Table of contents

File Operations.....	3
APT Package Management.....	4
C Compilation.....	5
Cleanup and Verification.....	7

List of Figures

Figure 1: Essential file commands.....	3
Figure 2: APT commands.....	4
Figure 3: Creating source file, compiling, and executing.....	5
Figure 4: Writing C program.....	5
Figure 5: Modifying and recompiling.....	6
Figure 6: Modifying C program.....	6
Figure 7: Final checks.....	7

File Operations

```
sailenawale@shriya-MacBook-Air-52 ~ % docker run -it ubuntu
root@0abec3649c1d:/# pwd
/
root@0abec3649c1d:/# ls
bin boot dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var
root@0abec3649c1d:/# mkdir ~/linux_lab
root@0abec3649c1d:/# ls
bin boot dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var
root@0abec3649c1d:/# cd home
root@0abec3649c1d:/home# ls
ubuntu
root@0abec3649c1d:/home# cd ~
root@0abec3649c1d:~# pwd
/root
root@0abec3649c1d:~# ls
linux_lab
root@0abec3649c1d:~# cd ~/linux_lab
root@0abec3649c1d:~/linux_lab# pwd
/root/linux_lab
```

```
root@0abec3649c1d:~/linux_lab# nano greeting.txt
root@0abec3649c1d:~/linux_lab# cat greeting.txt
"Hello Linux!"
root@0abec3649c1d:~/linux_lab# cp greeting.txt backup.txt
root@0abec3649c1d:~/linux_lab# mv backup.txt old_greeting.txt
root@0abec3649c1d:~/linux_lab# ls
greeting.txt  old_greeting.txt
root@0abec3649c1d:~/linux_lab# rm old_greeting.txt
root@0abec3649c1d:~/linux_lab# ls
greeting.txt
root@0abec3649c1d:~/linux_lab# mkdir temp_dir
root@0abec3649c1d:~/linux_lab# ls
greeting.txt  temp_dir
root@0abec3649c1d:~/linux_lab# rmdir temp_dir
root@0abec3649c1d:~/linux_lab# ls
greeting.txt
root@0abec3649c1d:~/linux_lab# ls -l greeting.txt
-rw-r--r-- 1 root root 15 Sep  9 10:46 greeting.txt
root@0abec3649c1d:~/linux_lab#
```

Figure 1: Essential file commands

Summary

File operations mainly focuses on creating and removing directories, creating and editing files, copying, renaming, and removing files. It also includes commands to view file contents and check file permissions.

APT Package Management

```
root@0abec3649c1d:~/linux_lab# apt update
Hit:1 http://ports.ubuntu.com/ubuntu-ports resolute InRelease
Hit:2 http://ports.ubuntu.com/ubuntu-ports resolute-updates InRelease
Hit:3 http://ports.ubuntu.com/ubuntu-ports resolute-backports InRelease
Hit:4 http://ports.ubuntu.com/ubuntu-ports resolute-security InRelease
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@0abec3649c1d:~/linux_lab# apt install build-essential
Upgrading:
  libbz2-1.0  libc-bin  libc-gconv-modules-extra  libc6  perl-base

Installing:
  build-essential

root@0abec3649c1d:~/linux_lab# gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

root@0abec3649c1d:~/linux_lab# apt search python3
astro-python3/resolute 5.0 all
  Python 3 packages for astronomy

berrynet-dashboard/resolute 3.10.2-2build1 all
  deep learning gateway - python3 modules

cantor-backend-python3/resolute 4:25.12.3-0ubuntu1 arm64
  Python3 backend for Cantor
```

Figure 2: APT commands

Summary

This section covers basic APT commands such as updating packages, installing build-essential (GCC), checking its version, and searching for available packages.

C Compilation

```
root@0abec3649c1d:~/linux_lab# nano hello.c
root@0abec3649c1d:~/linux_lab# gcc hello.c -o hello_program
root@0abec3649c1d:~/linux_lab# ls
greeting.txt  hello.c  hello_program
root@0abec3649c1d:~/linux_lab# ls -l hello_program
-rwxr-xr-x 1 root root 70496 Sep  9 10:58 hello_program
root@0abec3649c1d:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
root@0abec3649c1d:~/linux_lab# gcc -Wall hello.c -o hello_program_warn ./hello_program_warn
root@0abec3649c1d:~/linux_lab# ls
greeting.txt  hello.c  hello_program  hello_program_warn
root@0abec3649c1d:~/linux_lab#
```

Figure 3: Creating source file, compiling, and executing

```
GNU nano 8.7.1
#include <stdio.h>

int main(){
    int a = 10, b = 5;
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

Figure 4: Writing C program

Summary

This section focuses on writing a C program using nano editor. The program is saved, compiled, and run using GCC. It was further checked for warnings.

```
root@0abec3649c1d:~/linux_lab# nano hello.c
root@0abec3649c1d:~/linux_lab# gcc -Wall hello.c -o hello_program
root@0abec3649c1d:~/linux_lab# ls
greeting.txt  hello.c  hello_program  hello_program_warn
root@0abec3649c1d:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 20 and 5 is 25
root@0abec3649c1d:~/linux_lab#
```

Figure 5: Modifying and recompiling

```
#include <stdio.h>

int main(){
    int a = 20, b = 5;
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

Figure 6: Modifying C program

Summary

In this section, a change was made in C program, which was further saved, compiled, and run again to check the updated result.

Cleanup and Verification

```
root@0abec3649c1d:~/linux_lab# cd ~
root@0abec3649c1d:~# ls -lh ~/linux_lab/
total 40K
-rw-r--r-- 1 root root 15 Sep  9 10:46 greeting.txt
-rw-r--r-- 1 root root 146 Sep  9 11:00 hello.c
-rwxr-xr-x 1 root root 69K Sep  9 11:01 hello_program
-rwxr-xr-x 1 root root 69K Sep  9 10:59 hello_program_warn
root@0abec3649c1d:~# cat ~/linux_lab/hello.c
#include <stdio.h>

int main(){
    int a = 20, b=5;
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);
    return 0;
}
root@0abec3649c1d:~# rm -rf ~/linux_lab
root@0abec3649c1d:~# ls
root@0abec3649c1d:~#
```

Figure 7: Final checks

Summary

This section covers final checking on the files in the “linux_lab” directory, and also checking on the C program. Further, removing the directory once the work is done.